

Xuangeng CHU

xuangeng.chu@mi.t.u-tokyo.ac.jp | xg.chu@outlook.com

Research Interests

1. Novel view synthesis (NeRF / 3DGS).
2. Human-centric 3D vision, human shape / pose estimation, generation and reconstruction.

Education

The University of Tokyo Ph.D. Candidate in RCAST, University Fellowship, Supervisor: Prof. Tatsuya Harada	2023 – now Tokyo, Japan
Peking University M.Eng. in Software Engineering, Supervisor: Prof. Yasha Wang	2018 – 2021 Beijing, China
Tongji University B.Eng. in Computer Science	2014 – 2018 Shanghai, China

Work Experience

Tencent, Research Engineer Applied Research Center	Mar 2021 – Oct 2022 Shenzhen, China
• Worked on category extensible object detection algorithm for videos. • Worked on articulated model reconstruction algorithm for a virtual scene generating system.	

Internship Experience

International Digital Economy Academy, Research Intern Advised by Dr. Yu LI	Dec 2022 - Nov 2023 Shenzhen, China / Remote
<i>Worked on research and development of human reconstruction and pose estimation.</i>	
Microsoft Research Asia, Research Intern Advised by Dr. Xiulian PENG	Jun 2020 - Feb 2021 Beijing, China
<i>Worked on research of audio-visual speech separation problem (cocktail party problem).</i>	
MEGVII Technology, Research Intern Advised by Dr. Xiangyu ZHANG	Jan 2019 - Jun 2020 Beijing, China
<i>Worked on research and development of object detection in crowded scenes.</i>	

Publication

GPAvatar: Generalizable and Precise Head Avatar from Image(s) Xuangeng Chu , Yu Li, Ailing Zeng, Tianyu Yang, Lijian Lin, Yunfei Liu, Tatsuya Harada	ICLR 2024
<i>We propose a framework to reconstructs 3D head avatars from one or several images in a single forward pass. The key idea of this work is a dynamic point-based expression field and a attention-based fusion module. Project website and code: https://xg-chu.github.io/project_gpavatar.</i>	

Accurate 3D Face Reconstruction with Facial Component Tokens

ICCV 2023

Tianke Zhang, **Xuangeng Chu**, Yunfei Liu, Lijian Lin, Zhendong Yang, Zhengzhuo Xu, Chengkun Cao, Fei Yu, Changyin Zhou, Chun Yuan, Yu Li

We propose a framework for 3D face reconstruction from monocular images based on 3DMM and transformers. Our method uses separate tokens to improve the disentanglement of shape and expression for more accurate reconstruction.

Detection in Crowded Scenes: One Proposal, Multiple Predictions

CVPR 2020 Oral

Xuangeng Chu*, Anlin Zheng*, Xiangyu Zhang*, Jian Sun

We propose a simple and almost cost-free method to improve the detection performance in crowded scenes. The key idea of this work is to predict a set of instances from each proposal region instead of just one. Code is available on: <https://github.com/xg-chu/CrowdDet>.

Services

Reviewer, IEEE Conference on Computer Vision and Pattern Recognition (CVPR) 2022, 2023